

GYUMRI, Armenia

WMO: 376860

Lat: 40.763N

Lon: 43.856E

Elev: 4963

StdP: 12.24

Time Zone: 4.00 (E04)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-9.6	-5.5	-14.8	3.0	-9.9	-9.7	3.9	-4.9	21.3	25.0	18.0	26.6	2.0	40	0.600

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	25.8	89.5	61.6	86.2	61.6	83.8	60.7	64.8	81.9	63.5	80.3	62.1	78.6	9.0	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
59.0	89.7	73.9	57.3	84.3	70.9	55.7	79.4	69.5	32.8	81.6	31.6	80.5	30.6	78.5	69.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 24.6	21.5	18.5	DB	-10.9	93.5	9.4	3.7	-17.7	96.1	-23.2	98.3	-28.4	100.3	-35.3	103.0	(4)
(5)			WB	-11.5	67.1	9.2	1.6	-18.1	68.2	-23.5	69.1	-28.7	70.0	-35.3	71.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	46.0	18.6	22.0	34.9	46.0	54.0	62.6	69.9	70.6	61.7	49.7	36.3	24.4	(6)
(7)		DBStd	19.23	10.48	11.11	8.59	6.11	4.44	5.07	4.93	5.12	6.60	5.43	6.90	9.82	(7)
(8)		HDD50	3671	975	785	470	148	14	0	0	0	5	72	410	792	(8)
(9)		HDD65	7345	1440	1205	933	569	341	105	14	13	135	473	860	1257	(9)
(10)		CDD50	2224	0	0	2	30	138	377	618	640	355	64	0	0	(10)
(11)		CDD65	422	0	0	0	0	0	32	167	188	35	0	0	0	(11)
(12)		CDH74	4548	0	0	0	7	13	439	1669	1839	572	9	0	0	(12)
(13)		CDH80	1472	0	0	0	0	0	74	566	687	145	0	0	0	(13)
(14)	Wind	WSAvg	5.3	2.7	2.8	5.5	6.1	5.4	6.5	9.2	9.6	6.0	3.9	3.2	2.4	(14)
(15)	Precipitation	PrecAvg	18.5	0.9	1.0	1.1	2.2	3.2	2.8	1.8	1.3	1.0	1.5	1.0	0.9	(15)
(16)		PrecMax	31.2	1.9	2.4	2.8	4.8	5.4	6.4	4.3	3.2	2.8	4.4	2.2	2.1	(16)
(17)		PrecMin	13.7	0.0	0.1	0.0	0.2	1.2	0.9	0.2	0.0	0.0	0.2	0.1	0.0	(17)
(18)		PrecStd	3.5	0.5	0.6	0.6	1.0	1.1	1.4	1.1	0.7	0.7	0.9	0.6	0.6	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	41.3	48.6	62.8	73.3	75.5	86.0	93.1	94.6	88.1	75.2	59.4	51.4	(19)
(20)			MCWB	35.9	40.4	46.2	54.0	55.9	59.5	60.8	61.1	59.0	51.8	45.0	38.7	(20)
(21)		2%	DB	37.5	43.1	57.3	67.9	72.0	82.2	89.4	89.8	84.3	71.3	56.8	43.2	(21)
(22)			MCWB	33.2	37.0	43.8	50.0	54.4	59.6	62.9	61.1	57.4	51.4	43.6	35.4	(22)
(23)		5%	DB	35.4	39.4	53.3	64.1	69.6	78.8	86.1	86.4	80.5	67.6	52.2	39.4	(23)
(24)			MCWB	32.0	34.4	41.6	48.1	54.0	59.2	63.2	61.4	57.1	50.6	42.2	35.0	(24)
(25)		10%	DB	33.4	35.8	49.1	60.4	66.3	75.6	82.6	84.1	76.9	62.9	48.6	37.0	(25)
(26)			MCWB	31.3	32.3	39.7	46.2	52.3	58.6	62.0	60.9	56.0	49.4	40.7	33.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	37.5	41.3	48.2	56.3	59.0	64.1	67.6	66.7	62.1	54.9	47.7	40.5	(27)
(28)			MCD B	39.2	48.0	59.3	72.1	71.0	78.9	85.4	85.5	80.2	66.8	55.3	45.3	(28)
(29)		2%	WB	34.0	37.5	44.9	51.3	56.8	62.4	65.6	64.2	59.9	53.4	45.0	37.4	(29)
(30)			MCD B	35.4	42.3	55.4	63.8	68.0	77.4	82.9	82.6	78.1	65.6	51.8	41.1	(30)
(31)		5%	WB	32.4	34.8	42.4	49.0	55.2	61.0	64.3	62.8	58.4	52.0	43.6	35.0	(31)
(32)			MCD B	34.3	38.3	50.8	60.5	65.9	75.0	80.7	81.0	75.9	63.4	50.6	37.9	(32)
(33)		10%	WB	30.5	32.5	40.3	47.3	53.5	59.5	62.9	61.5	57.2	50.5	41.8	33.3	(33)
(34)			MCD B	32.8	35.2	47.5	57.9	63.9	73.1	79.1	79.2	73.4	61.2	48.2	35.7	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	18.0	20.3	21.3	23.7	22.9	25.3	25.3	25.8	27.8	24.8	22.9	18.2	(35)
(36)			MCDBR	15.8	19.5	27.5	30.1	28.1	28.4	28.5	28.9	31.9	32.6	28.8	21.9	(36)
(37)			MCWBR	12.1	13.8	15.5	14.7	13.8	11.7	10.0	9.4	11.1	15.8	17.9	15.4	(37)
(38)		5% WB	MCDBR	12.2	17.5	23.5	26.2	24.6	25.6	25.6	25.8	28.6	26.8	25.3	17.7	(38)
(39)			MCWBR	9.6	12.4	13.9	13.3	12.9	11.7	9.9	9.3	10.7	13.6	16.1	12.8	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.275	0.307	0.356	0.413	0.388	0.371	0.407	0.400	0.345	0.342	0.294	0.272	(40)	
(41)		taud	2.336	2.254	2.202	2.098	2.259	2.352	2.265	2.281	2.418	2.370	2.463	2.414	(41)	
(42)		Ebn at Noon	281	286	282	272	282	286	274	273	284	270	273	274	(42)	
(43)		Edh at Noon	27	35	41	49	43	39	42	40	33	31	24	23	(43)	
(44)	All-Sky Solar Radiation	RadAvg	740	1095	1341	1491	1759	2116	2137	1959	1584	1039	704	559	(44)	
(45)		RadStd	66	115	140	159	115	137	120	110	106	86	93	73	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (16 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page